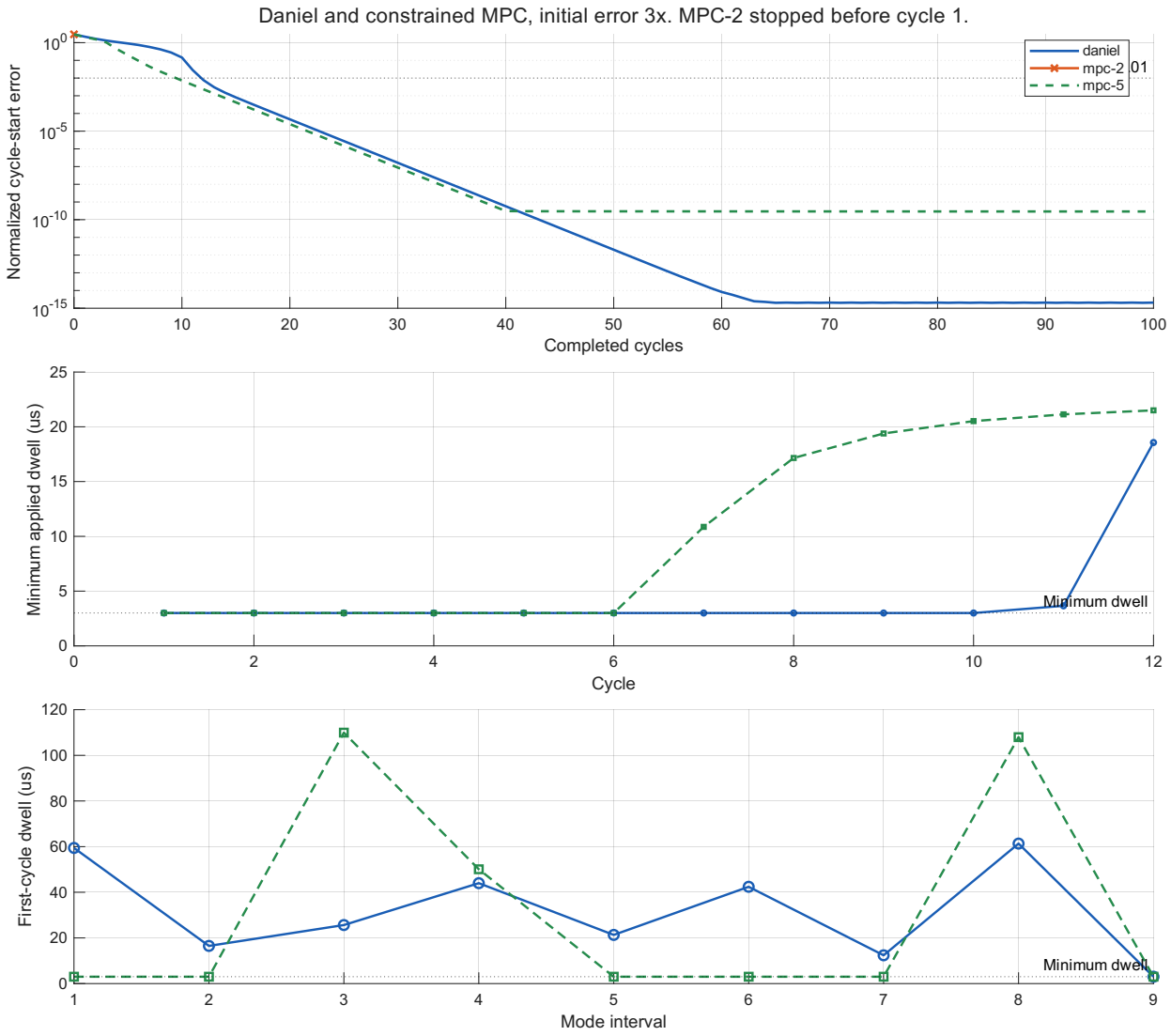
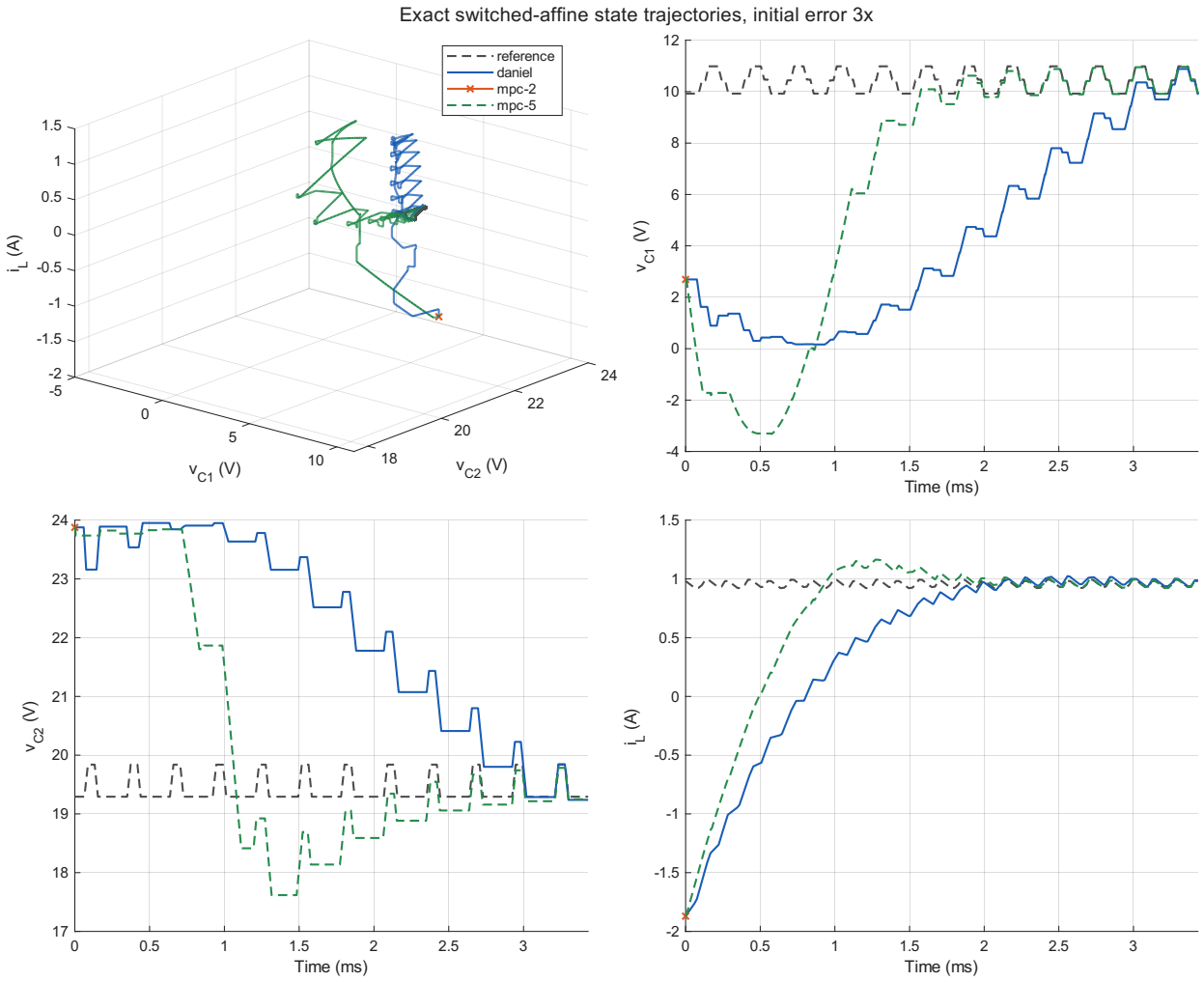
Daniel versus MPC

The threefold initial-error trial uses the same converter, LQR weights, 286 μs period, and 3 μs dwell bound. Blue denotes `daniel`, orange `mpc-2`, and dashed green `mpc-5`; the MPC suffix is its prediction horizon in cycles. Two-cycle MPC is infeasible at the first solve and applies no fallback. The panels show cycle-start error, minimum applied dwell, and first-cycle dwells.



Exact state trajectories over the first twelve nominal periods compare Daniel and five-cycle MPC with the dashed reference. Five-cycle MPC settles earlier but produces negative capacitor voltage. This trial also starts with negative current; it tests the mathematical model, not validated hardware operation. No nonlinear stability guarantee follows from these plots.



Computation time

Recorded MATLAB times per successful cycle include Daniel's LQR command and dwell conditioning, or MPC's QP solve. The table covers all three initial-error scales; each completed run supplies 100 calls. First-call overhead is included. Offline design, state normalization, plant propagation, and plotting are excluded. These are desktop timings, not ESP32 deadline measurements.

Scale	Method	Calls	Median (μs)	95th percentile (μs)
1x	daniel	100	13.25	131.20
1x	mpc-2	100	282.35	1830.45
1x	mpc-5	100	257.75	321.80
2x	daniel	100	11.15	16.65
2x	mpc-2	100	189.70	306.50
2x	mpc-5	100	303.35	478.20
3x	daniel	100	11.20	16.65
3x	mpc-2	0	–	–
3x	mpc-5	100	316.85	532.80

At 3x, the single infeasible MPC solve took 1525.10 μs; it is excluded from the successful-call summaries.

The 95th percentile describes the slower tail of the recorded calls, not a worst-case execution bound. Missing entries mean no successful MPC calls; the failed solve is reported separately below the table.